

EXP 4 Design of Low Pass Filter using Windowing Techniques

4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

PROGRAM

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000;    % Passband cutoff (Hz)

fs = 4000;    % Sampling frequency (Hz)

N = 50;      % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann);

%% --- Plot Frequency Responses ---

figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');
```

OUTPUT

```
-- Hanning Window Coefficients ---
0.0000
0
-0.0004
0
0.0013
0
-0.0028
0
0.0050
0
```

-0.0081
0
0.0122
0
-0.0179
0
0.0259
0
-0.0378
0
0.0580
0
-0.1027
0
0.3172
0.5000
0.3172
0
-0.1027
0
0.0580
0
-0.0378
0
0.0259
0
-0.0179
0
0.0122
0
-0.0081
0
0.0050
0
-0.0028
0
0.0013
0
-0.0004
0
0.0000

